Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

At a glance: Build a surface that sheds water and dirt the way a lotus leaf does. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: hydrophobicity, surface roughness, biomimicry, materials engineering (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- wax paper: about 2 sheets per team per run (1 roll for the camp)
- cardstock: about 3 sheets per team per run (16 total; from the shared cardstock stock)
- sandpaper strips: medium grit, about 2 per team plus spares (12 total)
- masking tape: to fix surfaces to the ramp (shared roll)
- pipettes: 3 mL graduated, 1 to 2 per team (one pack of 100 covers the camp; reused)
- water: tap, a few mL per droplet (fill jug plus pour-off bucket)
- pepper or cocoa: washable dirt proxy, a pinch per run (1 small shaker each)
- candle or paraffin wax: rubbed on cold to coat a surface (1 to 2 candles, shared)
- ramps: 1 per team at a fixed standard tilt (on-hand rigid boards)

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready, including a ramp built from an on-hand rigid board.

02 Set a fixed standard tilt for every ramp (a block or book under one edge, or a taped angle mark) so all teams run at the same slope, fill a water jug and stage a pour-off bucket, and put out the scoring sheet before testing begins.

03 Put out a containment tray at each station and have gloves available for students who do not want pepper, cocoa, or wax on their hands. No goggles are needed: this is dropping water onto a tilted surface over a tray, not a chemical or splash hazard.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook
0:00 - 0:05

Pose the mission as a challenge: "Build a surface that sheds water and dirt the way a lotus leaf does." Take a quick prediction by show of hands.

Phase 2 Prediction
0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints
0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve
0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams set the ramp to the standard tilt, release a measured droplet from the top, time the runoff over the tray, and score the residue left behind; they log this baseline before changing anything.

Phase 6 Redesign

Each team changes exactly one variable (texture, amount of wax, tilt, or surface material), predicts the effect, and re-runs at the same tilt with the same droplet and dirt amount to compare against baseline.

Phase 7 Score, debrief, cleanup
last 12 min

Teams submit a score slip, answer a debrief question with evidence, then reset the station: bin the used paper surfaces and dirt-laden wax paper, pour off standing water, wipe the ramp and tray clear of grit and dirt, dry the table, and wash hands (remove gloves first if worn).

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: The droplet soaks in or will not roll on the first run. Fix: Confirm the surface has both features (a rough texture and a wax coating, not one alone), check the ramp is at the standard tilt and not too shallow, use a smaller droplet, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to hydrophobic micro-texture.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES AskNature, Surface allows self-cleaning (lotus leaf); Koch and Barthlott, superhydrophobicity of the lotus leaf (contact angle about 163 degrees, hierarchical micro-papillae plus nano wax tubules).